How-To Library

Informatica

Performance Tuning Guidelines for PowerExchange for Microsoft Azure SQL Data Warehouse on the Spark Engine

© Copyright Informatica LLC 2018, 2023. Informatica, the Informatica logo, and Big Data Management are trademarks or registered trademarks of Informatica LLC in the United States and many jurisdictions throughout the world. A current list of Informatica trademarks is available on the web at https://www.informatica.com/trademarks.html

Abstract

When you use Informatica PowerExchange for Microsoft Azure SQL Data Warehouse to read data from or write data to Microsoft Azure SQL Data Warehouse, multiple factors such as hardware parameters, database parameters, application server parameters, and Informatica mapping parameters impact the adapter performance. You can optimize the performance by tuning these parameters appropriately. This article describes general reference guidelines to help you tune the performance of PowerExchange for Microsoft Azure SQL Data Warehouse on the Spark engine.

Supported Versions

- PowerExchange for Microsoft Azure SQL Data Warehouse 10.2.2

Table of Contents

Overview. 2
Performance Tuning Parameters. 3
Tune the Hardware. 3
Cluster Configuration based on Data Set. 3
Tune Microsoft Azure SQL Data Warehouse. 5
Tune the Spark Engine. 6
Tune the Mapping. 7
Scaling Data Warehouse Units. 7
Concurrent Connections to Microsoft Azure Blob Store. 7

Overview

Performance tuning is an iterative process in which you analyze the performance, use guidelines to estimate and define parameters that impact the performance and monitor and adjust the results as required.

This document describes the key hardware, database, Hadoop cluster, and Informatica mapping parameters that you can tune to optimize the performance of PowerExchange for Microsoft Azure SQL Data Warehouse for Spark. The document also includes case studies that involve loading data into Azure SQL Data Warehouse from HDFS and reading data from Azure SQL Data Warehouse to HDFS using Azure HDInsight 3.6.

Azure HDInsight simplifies running Hadoop and related big data applications on Azure. It removes the cost and complexity of managing the Hadoop installation. This means any developer or business has the power to do analytics without large capital expenditures. Today, you can spin up a performance-optimized Hadoop cluster in the Azure cloud within minutes on the latest high-performance computing hardware and network without making a capital investment to purchase the hardware. You can expand and shrink a running cluster on demand. This means if you need answers to your questions faster, you can immediately scale up the size of your cluster to crunch the data more quickly.

The Azure HDInsight and SQL Data Warehouse best practices are as follows:

- Host Informatica Domain on an Azure Virtual machine for better MRS Monitoring.
- Choose the SQL Data Warehouse in the region where the Azure HDInsight is configured, or choose the Azure HDInsight in the region where the SQL Data Warehouse is configured.
- Make sure that all the Spark level tuning is done. For details, see "Tune the Spark Engine" on page 6.
- Pick the right instance size for HDInsight nodes and Azure SQL Data Warehouse units. For details, see "Tune Microsoft Azure SQL Data Warehouse" on page 5.

Note: The performance testing results listed in this article are based on observations in an internal Informatica environment using data from real-world scenarios. The performance of PowerExchange for Microsoft Azure SQL Data Warehouse might vary based on individual environments and other parameters even when you use the same data.

- It is preferable to use Microsoft Azure SQL Data Warehouse Gen2 over Microsoft Azure SQL Data Warehouse Gen1 for better performance.

Performance Tuning Parameters

You can optimize the performance of Microsoft Azure SQL Data Warehouse mappings by tuning the following areas:

- Tune the Hardware
- Tune the Spark Engine
- Tune the Mapping

Tune the Hardware

To optimize the performance, acquire the right type of hardware for your Big Data Management environment.

Cluster Configuration based on Data Set

You can tune the hardware by configuring the Hadoop cluster based on different data sets.

Default Data Set

A default data set represents a data set that is processed with all default settings in given amount of time. With default settings, a two head node and eight worker node cluster can load 7.5 GB of data from an HDFS data source to an Azure SQL Data Warehouse in 00:09:11 (hh:mm:ss) and from an Azure SQL Data Warehouse to an HDFS data source in 00:11:13 (h:mm:ss) when you run mappings on the Spark engine.

The following table shows the sizing outcome for default settings:

	Infrastructure	**Hadoop Services**
Services	Domain, MRS for Monitoring, DIS	Yarn Resource Manager, Node Manager
vCPUs	2	(4*2) + (8*2) = 24
Memory	8 GB	(7*2) + (28 *2) = 70 GB
Storage	14 GB	75 * 3 = 225 GB

Note: Total vCPUs = vCPUs of 1 Master + (vCPUs in core Node * No of Core Nodes).

Total Memory = (Master Node Process Memory + (Node Process Memory * Number of core Nodes)).

Total Storage = (I/P Data * Replication Factor)

Sandbox Data Set

Sandbox data set is defined as a system that loads 1 GB of data from an HDFS data source to an Azure SQL Data Warehouse and from an Azure SQL Data Warehouse to an HDFS data source in when you run mappings on the Spark engine.

The following table shows the sizing outcome for a Sandbox data set:

	Infrastructure	Hadoop Services
Services	Domain, MRS for Monitoring, DIS	Yarn Resource Manager, Node Manager
vCPUs	2	(4*2) + (8*1) = 16
Memory	8 GB	(4*2) + (7*1) = 15 GB
Storage	14 GB	1*2 = 2 GB

Small Data Set

Small data set is defined as a system that loads 7.5 GB of data from an HDFS data source to an Azure SQL Data Warehouse in 00:09:11 (hh:mm:ss) and from an Azure SQL Data Warehouse to an HDFS data source in 00:11:13 (hh:mm:ss) hour when you run mappings on the Spark engine.

The following table shows the sizing recommendations for a small data set:

	Infrastructure	Hadoop Services
Services	Domain, MRS for Monitoring, DIS	Yarn Resource Manager, Node Manager
vCPUs	2	(4*2) + (4*1) = 12
Memory	8 GB	(2*2) + (2*1) = 6 GB
Storage	14 GB	7.5*3 = 22.5 GB

Medium Data Set

Medium data set is defined as a system that loads 75 GB of data from an HDFS data source to an Azure SQL Data Warehouse 01:10:03 and from an Azure SQL Data Warehouse to an HDFS data source in 00:36:33 hour when you run mappings on the Spark engine.

The following table shows the sizing recommendations for a medium data set:

	Infrastructure	Hadoop Services
Services	Domain, MRS for Monitoring, DIS	Yarn Resource Manager, Node Manager
vCPUs	2	(4*2) + (8*4) = 40
Memory	8 GB	(4*2) + (7 *4) = 36 GB
Storage	14 GB	75 * 3 = 225 GB

Large Data Set

A Large data set is defined as a system that loads 750 GB of data from an HDFS data source to an Azure SQL Data Warehouse storage in 02:30:00 (hh:mm:ss) hour when you run mappings on the Spark engine.

The following table shows the sizing recommendation for a large data set:

	Infrastructure	Hadoop Services
Services	Domain, MRS for Monitoring, DIS	Yarn Resource Manager, Node Manager
vCPUs	2	(4*2) + (8*45) = 368
Memory	8 GB	(7*2) + (7 *45) = 329GB
Storage	14 GB	750 * 3 = 2.25 TB

Tune Microsoft Azure SQL Data Warehouse

The following table describes the tuning recommendations for Microsoft Azure SQL Data Warehouse:

Parameter	Default Data Set	Sandbox Data Set	Small Data Set	Medium Data Set	Large Data Set
Number of Blob Connections	4	4	4	10	10
SQL Data Warehouse Units	DW100	DW100	DW100	DW350 or above	DW1000 or above

You can tune the following Microsoft Azure SQL Data Warehouse components to improve the read and write performance:

Virtual Machine Size

For performance sensitive applications, you can use the following virtual machines sizes for Microsoft Azure SQL Data Warehouse:

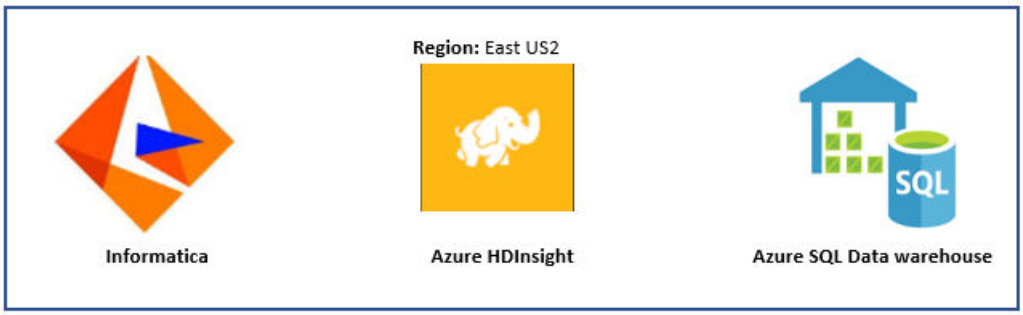
- Microsoft SQL Server Enterprise Edition DS3 or later
- Microsoft SQL Server Standard and Web Editions DS2 or later

For recommendations by Microsoft Azure for the virtual machine sizes and options that you can use to optimize performance, see https://docs.microsoft.com/en-us/azure/virtual-machines/windows/sizes?toc=%2Fazure%2Fvirtual-machines%2Fwindows%2Ftoc.json

Storage

You must create a Microsoft Azure storage account in the same data center as the Microsoft SQL Server virtual machines to reduce transfer delays. When you create a storage account, disable geo-replication as consistent write order across multiple disks is not a guarantee.

The following image shows the Microsoft Azure storage account and virtual machine in the same data center:



When the domain and the storage accounts are at different data centers, additional network-related data transfer delays might occur. The delays might increase substantially as the data volume increases.

As standard storage has varying latencies and bandwidth, you can use premium storage of virtual machines of DS-series along with DSv2-series and GS-series for optimal performance.

For more information about optimizing your design with Premium Storage, see https://docs.microsoft.com/en-us/azure/virtual-machines/windows/premium-storage.

Disks

Consider the following recommendations when you use disks:

- Use a minimum of two disks, one for the log files and the other for the data files and the TempDB. For more information about improving the speed and performance of the disks, see https://docs.microsoft.com/en-us/azure/virtual-machines/windows/premium-storage#scalability-and-performance-targets-when-using-premium-storage.
- Do not use the operating system or temporary disks for database storage or logging.
- Enable read caching on the disks that host the data files and TempDB.

Tune the Spark Engine

The following table lists the tuning recommendations for the Spark parameters:

Parameter	Default Value	Sandbox Dataset Recommendation	Small Dataset Recommendation	Medium Dataset Recommendation	Large Dataset Recommendation
spark.executor.memory	6 GB	1 GB	1 GB	1 GB	1 GB
spark.executor.cores	2	1	1	1	1
spark.executor.instances	10	10	3	30	300
spark.driver.memory	4 GB	1 GB	2 GB	2 GB	4 GB

The following list describes the Spark tuning parameters:

- spark.executor.memory. The memory to use per executor process.

  Note: If you have passthrough mappings, use the recommended spark parameters. For more information about the Spark parameters, see *BDM Performance Tuning Guide*.
- spark.executor.cores. The number of cores to use on each executor.

- spark.executor.instances. The total number of executors to start. This parameter is honored only when spark.shuffle.service.enabled and spark.dynamicallocation.enabled are set to false. If the spark shuffle service is enabled, spark dynamic allocation is enabled by default and the value provided for this property is not honored.
- spark.driver.memory. The memory to use for the driver process where SparkContext is initialized.

Note: Any changes to the Spark parameters must be done in the cluster configuration object. You need not restart Data Integration Service post the changes. The changes are expected to be picked automatically for the subsequent runs after the changes.

- In a small, medium, or large data set, you might get the following error:

  ```
  Container is running beyond physical memory limits.
  ```

  To resolve this error, increase the value of `spark.executor.memory` parameter from 1GB to 2GB.

- In a default data set, you might get the following error:

  ```
  java.lang.IllegalArgumentException: Required executor memory (6144+614 MB) is above the max threshold (6144 MB) of this cluster! Please check the values of 'yarn.scheduler.maximum-allocation-mb' and/or 'yarn.nodemanager.resource.memory-mb.
  ```

  To resolve this error, decrease the value of `spark.executor.memory` parameter from 6GB to 4GB. Otherwise, increase the value of `yarn.scheduler.maximum-allocation-mb` to 10GB.

Tune the Mapping

When you run mappings on the Spark engine, consider the following tuning recommendations and best practices.

Scaling Data Warehouse Units

The query performance is directly proportional to the size of the Azure SQL Data Warehouse Units (DWU). Higher the size, higher the query slots, and higher the number of parallel queries that can run in parallel.

For more information on DWU, see https://docs.microsoft.com/en-us/azure/sql-data-warehouse/what-is-a-data-warehouse-unit-dwu-cdwu and https://docs.microsoft.com/en-us/azure/synapse-analytics/sql-data-warehouse/memory-concurrency-limits.

Concurrent Connections to Microsoft Azure Blob Store

When you read data from or write data to Microsoft Azure SQL Data Warehouse, you can specify the number of concurrent connections to the Microsoft Azure Blob Storage.

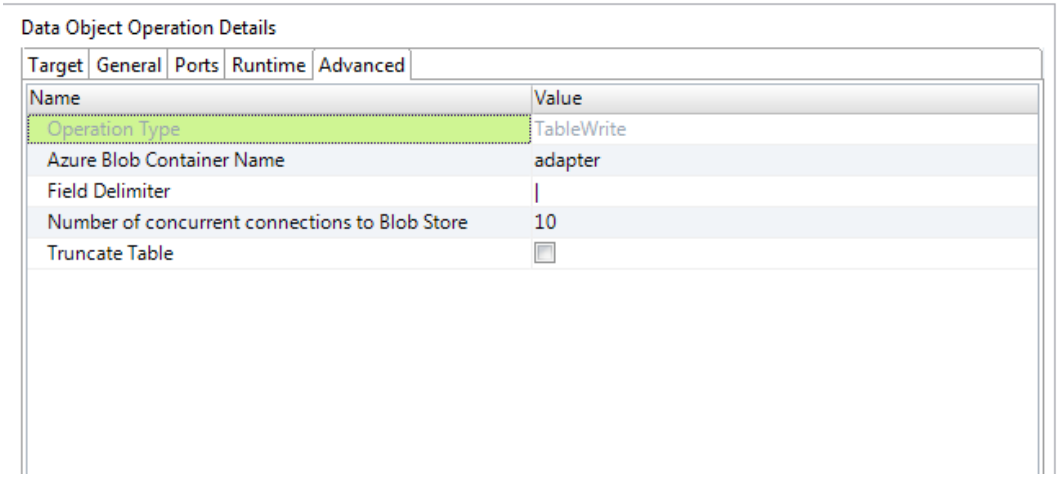
In a read operation, you can specify the number of concurrent connections to the Microsoft Azure Blob Store to download files. In a write operation, you can specify the number of concurrent connections to the Microsoft Azure Blob Store to upload files. The default number of concurrent connections set for a Microsoft Azure SQL Data Warehouse object is 4.

You can improve the performance of reading or writing data to Microsoft Azure SQL Data Warehouse by increasing the number of concurrent connections to the Microsoft Azure Blob Store.

To increase the number of concurrent connections in Informatica Developer, perform the following steps:

1. Log in to Informatica Developer.
2. Double-click and open the Microsoft Azure SQL Data Warehouse data object read operation.
3. Click the **Advanced** tab.

4. Increase the number of concurrent connections to the Microsoft Azure Blob Store for the TableWrite and TableRead operation type.
   The following image displays the advanced properties for the Microsoft Azure Blob Storage data object write operation in the **Advanced** tab of the Developer tool:



Author